

1. DATASET GENERATION

INPUT

ICF code combinations

- 38 combinations Brief Core Set generalized chronic pain

GENERATOR FLOW (n8n + Gemini)

Automated in silico generation

- Medical prompting with Self-Verification
- 3 clinical narratives per combination

OUTPUT

114 Synthetic clinical narratives

- Structured clinical free-text

2. AUTOMATED CODING

INPUT

Unstructured clinical text

- 114 clinical cases (blinded, no tags/IDs)

CODIFIER FLOW (n8n + LLMs)

Automated clinical coding

- Multi-model inference
- 3 iterations per case + consensus rule

OUTPUT

Coded clinical dataset

- Predicted ICF codes per run and consensus

3. RESULTS & BENCHMARK

INPUT

Predictions vs Ground Truth

- Matrices of target vs predicted codes

ANALYSIS SCRIPTS (Python & R)

Statistical evaluation

- MultiLabel binarization & prevalence
- Bootstrapping for confidence intervals

OUTPUT

Comparative results

- F1-score (Micro/Macro), PAE
- Gwet's AC1 and Krippendorff's α